

Haplotyping as Perfect Phylogeny: A Direct Approach

Vineet Bafna, Dan Gusfield, Giuseppe Lancia and Shibu Yooseph

Journal of Computational Biology, Vol. 10, No. 3, pp. 323-340, 2003

Abstract

A full haplotype map of the human genome will prove extremely valuable as it will be used in large-scale screens of populations to associate specific haplotypes with specific complex genetic-influenced diseases. A haplotype map project has been announced by NIH. The biological key to that project is the surprising fact that some human genomic DNA can be partitioned into long blocks where genetic recombination has been rare, leading to strikingly fewer distinct haplotypes in the population than previously expected (Helmuth, 2001; Daly et al., 2001; Stephens et al., 2001; Friss et al., 2001). In this paper we explore the algorithmic implications of the no-recombination in long blocks observation, for the problem of inferring haplotypes in populations.

Abstract (Contd.)

This assumption, together with the standard population-genetic assumption of infinite sites, motivates a model of haplotype evolution where the haplotypes in a population are assumed to evolve along a coalescent, which as a rooted tree is a perfect phylogeny. We consider the following algorithmic problem, called the perfect phylogeny haplotyping problem (PPH), which was introduced by Gusfield (2002) - given n genotypes of length m each, does there exist a set of at most $2n$ haplotypes such that each genotype is generated by a pair of haplotypes from this set, and such that this set can be derived on a perfect phylogeny? The approach taken by Gusfield (2002) to solve this problem reduces it to established, deep results and algorithms from matroid and graph theory. Although that reduction is quite simple and the resulting algorithm nearly optimal in speed, taken as a whole that approach is quite involved, and in particular, challenging to program.

Abstract (Contd.)

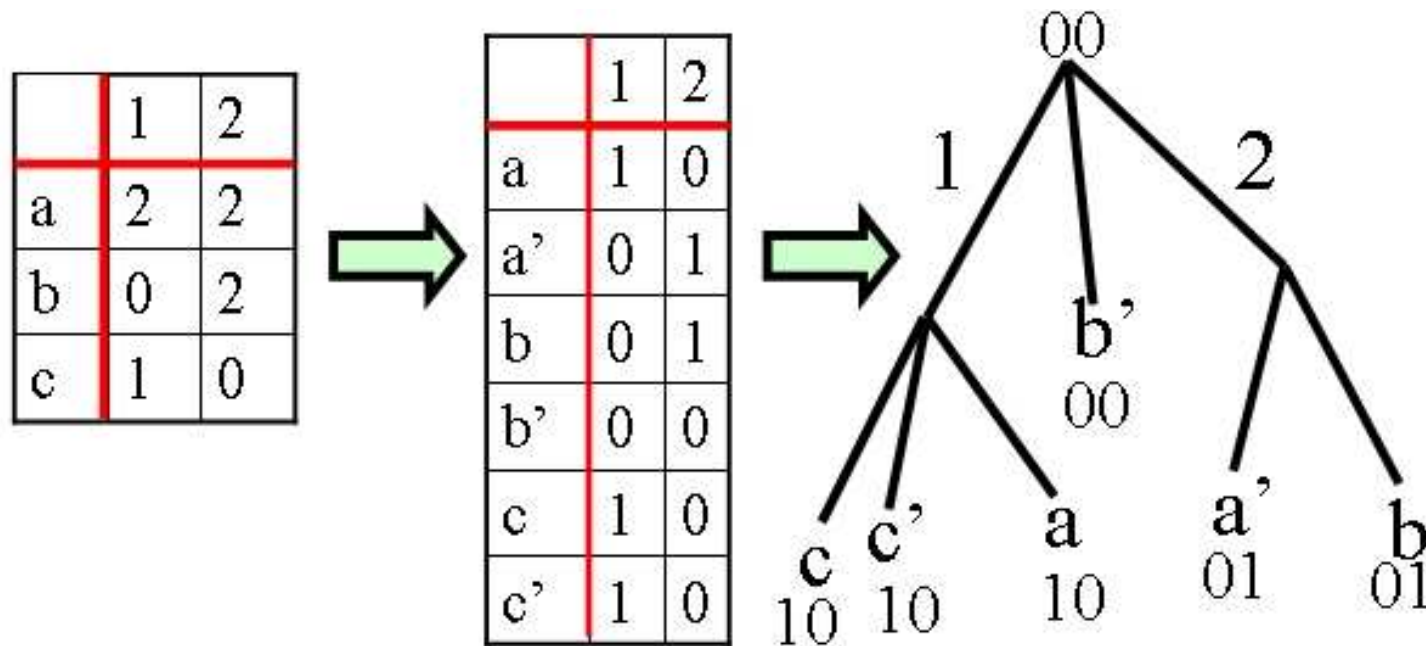
Moreover, anyone wishing to fully establish, by reading existing literature, the correctness of the entire algorithm would need to read several deep and difficult papers in graph and matroid theory. However, as stated by Gusfield (2002), many simplifications are possible and the list of "future work" in Gusfield (2002) began with the task of developing a simpler, more direct, yet still efficient algorithm. This paper accomplishes that goal, for both the rooted and unrooted PPH problems. It establishes a simple, easy-to-program, $O(nm^2)$ -time algorithm that determines whether there is a PPH solution for input genotypes and produces a linear-space data structure to represent all of the solutions. The approach allows complete, self-contained proofs.

Abstract (Contd.)

In addition to algorithmic simplicity, the approach here makes the representation of all solutions more intuitive than in Gusfield (2002), and solves another goal from that paper, namely, to prove a nontrivial upper bound on the number of PPH solutions, showing that that number is vastly smaller than the number of haplotype solutions (each solution being a set of n pairs of haplotypes that can generate the genotypes) when the perfect phylogeny requirement is not imposed.

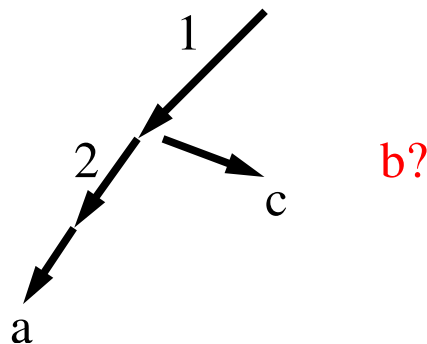
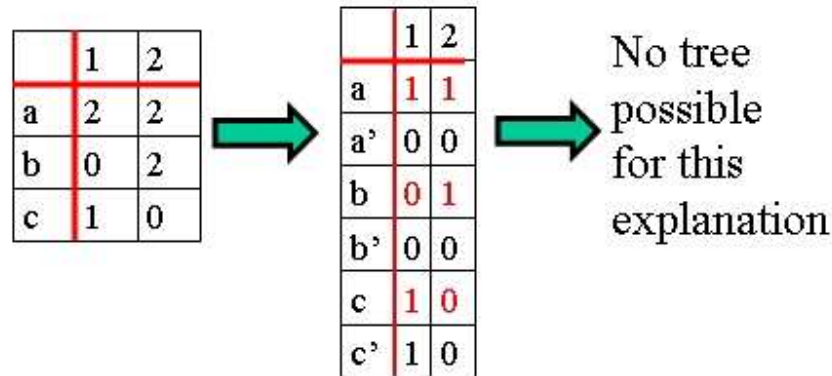
The Problem

- The Haplotype Phylogeny Problem: Given a set S of n genotype vectors, find an explaining set of haplotypes that fits a perfect phylogeny.



The Classical Theorem of Perfect Phylogeny

- A binary matrix B has a perfect phylogeny if and only if for each pair of columns, there are no three rows with values 0, 1; 1, 0; and 1, 1 in those two columns.



Observations

- Forced pattern: *equated* or *negated*:

	1	2	3
A	2	2	2
B	1	2	0
C	2	0	2

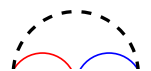
	1	2	3
A	2	2	2
B	1	2	0
C	2	0	2

- Inferred color:

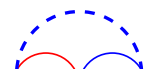
	1	2	3
A	2	2	2
B	1	2	0
C	2	0	2

Observations (Contd.)

■ How to infer color?



	1	2	3
A	2	2	2
B	1	2	0
C	2	0	2



	1	2	3
A	2	2	2
B	1	2	0
C	2	0	2

■ More examples:



	1	2	3
A	2	2	2
B	0	2	0
C	2	0	2



	1	2	3
A	2	2	2
B	0	2	0
C	2	0	2



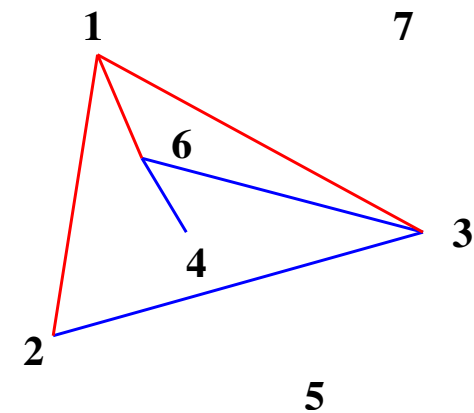
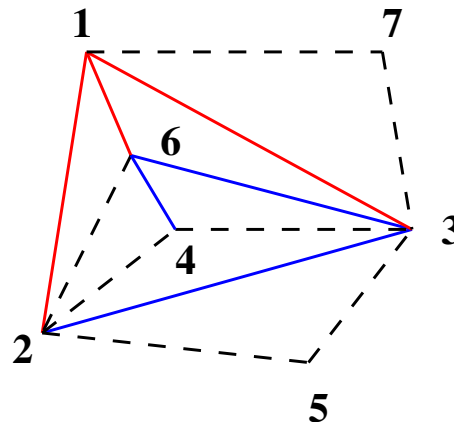
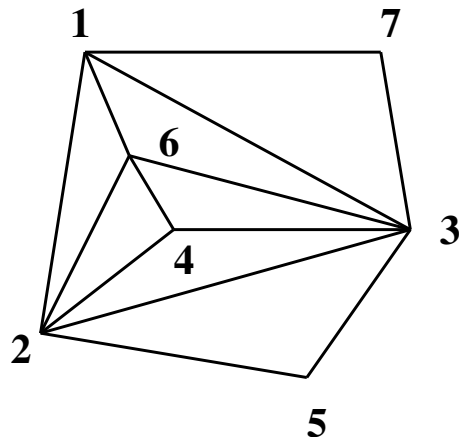
	1	2	3
A	2	2	2
B	1	2	1
C	0	2	1

An Example

■ The matrix M :

	1	2	3	4	5	6	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	0	2	0
c'	1	2	2	2	0	2	0
d	1	2	2	0	2	0	0
d'	1	2	2	0	2	0	0
e	2	2	0	0	0	2	0
e'	2	2	0	0	0	2	0

■ Graph G , G_c and G_f :



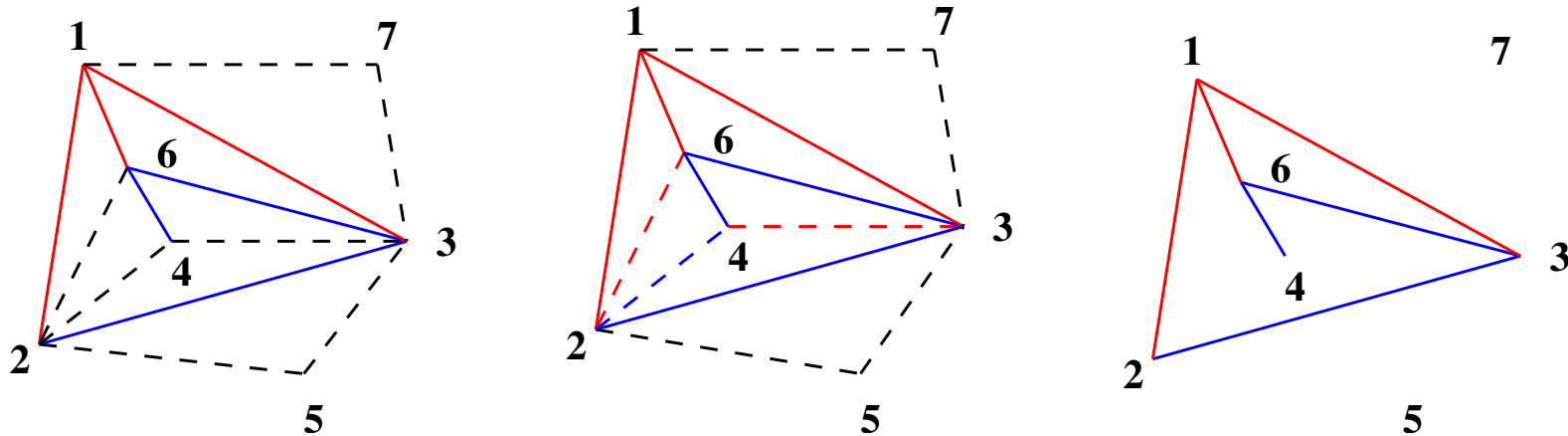
The Central Theorem

There is a solution to the PPH problem for M if and only if there is a coloring of the dashed edges of G_c with the following property:

For any triangle (i, j, k) in G_c , where there is one row containing 2's in all three columns i , j , and k , the coloring makes either 0 or 2 of the edges blue (*negated*).

Corollary

- Redo the example.



- Corollary 1: In a single connected component C of G with k connected components in G_f , there are exactly 2^{k-1} different solutions to the PPH problem in the columns of M represented by C .
- Corollary 2: If G has r connected components and t connected components of G_f , then there are exactly 2^{t-r} solutions to the PPH problem.

Redo the Example



	1	2	3	4	5	6	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	0	2	0
c'	1	2	2	2	0	2	0
d	1	2	2	0	2	0	0
d'	1	2	2	0	2	0	0
e	2	2	0	0	0	2	0
e'	2	2	0	0	0	2	0

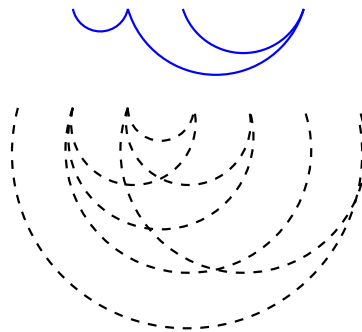
{{1, 2}, {1, 3}, (1, 6)}



	1	2	3	4	5	6	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	0	2	0
c'	1	2	2	2	0	2	0
d	1	2	2	0	2	0	0
d'	1	2	2	0	2	0	0
e	2	2	0	0	0	2	0
e'	2	2	0	0	0	2	0

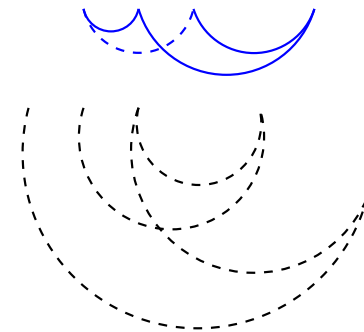
{{1, 2}, {1, 3}, (1, 6)}

{{2, 6}, {3, 4}}



{{2, 3}, {4, 6}, {3, 6}}

{{2, 5}, {3, 5}}
 {{1, 7}, {3, 7}}
 {{2, 4}, {2, 6}, {3, 4}}



{{2, 3}, {4, 6}, {3, 6}}

{{2, 4}}

{{2, 5}, {3, 5}}
 {{1, 7}, {3, 7}}

Redo the Example (Contd.)

	1	2	3	4	5	6	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	0	2	0
c'	1	2	2	2	0	2	0
d	1	2	2	0	2	0	0
d'	1	2	2	0	2	0	0
e	2	2	0	0	0	2	0
e'	2	2	0	0	0	2	0

→

	1	2	3	4	6	5	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	2	0	0
c'	1	2	2	2	2	0	0
d	1	2	2	0	2	0	0
d'	1	2	2	0	2	0	0
e	2	2	0	0	2	0	0
e'	2	2	0	0	2	0	0

→

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	0	0	0	0	0	0	2
b'	0	0	0	0	0	0	2
c	1	1	0	0	1	0	0
c'	1	0	1	0	0	0	0
d	1	1	0	0	0	2	0
d'	1	0	1	0	0	0	0
e	0	0	0	0	0	0	0
e'	0	0	0	0	0	0	0

The Solutions

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	2
b'	0	0	0	0	0	0	2
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	2	0
d'	1	0	1	0	0	2	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

→

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	1
b'	0	0	0	0	0	0	0
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	1	0
d'	1	0	1	0	0	0	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	1
b'	0	0	0	0	0	0	0
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	0	0
d'	1	1	0	0	0	0	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	0
b'	0	0	0	0	0	0	1
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	1	0
d'	1	0	1	0	0	0	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	0
b'	0	0	0	0	0	0	1
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	0	0
d'	1	1	0	0	0	0	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

The Algorithm

1. Find all forced relationships by considering columns in pairs.
2. Find all inferred relationships.
3. Use relationships to produce haplotype matrices

The Time Complexity

step 1: $O(nm^2)$

step 2: $O(m^2)$

step 3: $O(nm)$

$\Rightarrow O(nm^2)$

Thanks

Thanks

Date: Nov. 19, 2004

Created by: Liang-Tai Wei